

Q1 - 25 July - Shift 2

If the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ meets the line

$\frac{x}{7} + \frac{y}{2\sqrt{6}} = 1$ on the x-axis and the line

$\frac{x}{7} - \frac{y}{2\sqrt{6}} = 1$ on the y-axis, then the eccentricity

of the ellipse is

(A) $\frac{5}{7}$ (B) $\frac{2\sqrt{6}}{7}$

(C) $\frac{3}{7}$ (D) $\frac{2\sqrt{5}}{7}$

Space for your notes:

Q2 - 26 July - Shift 2

The acute angle between the pair of tangents drawn to the ellipse $2x^2 + 3y^2 = 5$ from the point (1,3) is

(A) $\tan^{-1}\left(\frac{16}{7\sqrt{5}}\right)$ (B) $\tan^{-1}\left(\frac{24}{7\sqrt{5}}\right)$

(C) $\tan^{-1}\left(\frac{32}{7\sqrt{5}}\right)$ (D) $\tan^{-1}\left(\frac{3+8\sqrt{5}}{35}\right)$

Space for your notes:

Q3 - 27 July - Shift 1

If the length of the latus rectum of the ellipse $x^2 + 4y^2 + 2x + 8y - \lambda = 0$ is 4, and l is the length of its major axis, then $\lambda + l$ is equal to _____.

Space for your notes:

Q4 - 28 July - Shift 2

Let the tangents at the points P and Q on the ellipse

$$\frac{x^2}{2} + \frac{y^2}{4} = 1 \text{ meet at the point } R(\sqrt{2}, 2\sqrt{2} - 2).$$

If S is the focus of the ellipse on its negative major axis, then $SP^2 + SQ^2$ is equal to

Space for your notes:

Q5 - 29 July - Shift 2

Let

$$S = \{(x, y) \in \mathbb{N} \times \mathbb{N} : 9(x - 3)^2 + 16(y - 4)^2 \leq 144\}$$

$$\text{and } T = \{(x, y) \in \mathbb{R} \times \mathbb{R} : (x - 7)^2 + (y - 4)^2 \leq 36\}.$$

The $n(S \cap T)$ is equal to _____.

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Answer Key

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Q1 (A)

Q2 (B)

Q3 (75)

Q4 (13)

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Q5 (27)

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Q1 (A)

(A)

Q2 (B)

Equation of tangent to the ellipse $2x^2 + 3y^2 = 5$ is

$$y = mx \pm \sqrt{\frac{5}{2}m^2 + \frac{5}{3}}$$

It pass through (1, 3)

$$3 = m \pm \sqrt{\frac{5}{2}m^2 + \frac{5}{3}}$$

$$3m^2 + 12m - \frac{44}{3} = 0$$

Let θ be the angle between the tangents

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan \theta = \left| \frac{3\sqrt{320}}{-35} \right|$$

$$\theta = \tan^{-1} \left(\frac{24}{7\sqrt{5}} \right)$$

Q3 (75)

$$\lambda + \ell = 75$$

$$x^2 + 4y^2 + 2x + 8y - \lambda = 0$$

$$\frac{(x+1)^2}{\lambda+5} + \frac{(y+1)^2}{\frac{\lambda+5}{4}} = 1$$

$$\therefore \frac{2b^2}{a} = 4$$

$$\frac{2(\lambda+5)}{4} = 4(\sqrt{\lambda+5})$$

$$\Rightarrow \lambda = 59$$

$$\lambda \neq -5$$

$$l = 2a = 2\sqrt{\lambda+5} = 2\sqrt{65} = 16$$

$$\Rightarrow \lambda + \ell = 59 + 16 = 75$$

Q4 (13)

Ellipse is

$$\frac{x^2}{2} + \frac{y^2}{4} = 1; e = \frac{1}{\sqrt{2}}; S \equiv (0, -\sqrt{2})$$

Chord of contact is

$$\frac{x}{\sqrt{2}} + \frac{(2\sqrt{2} - 2)y}{4} = 1$$

$$\Rightarrow \frac{x}{\sqrt{2}} = 1 - \frac{(\sqrt{2} - 1)y}{2} \text{ solving with ellipse}$$

$$\Rightarrow y = 0, \sqrt{2} \therefore x = \sqrt{2}, 1$$

$$P \equiv (1, \sqrt{2}) \quad Q \equiv (\sqrt{2}, 0)$$

$$\therefore (SP)^2 + (SQ)^2 = 13$$

Q5 (27)

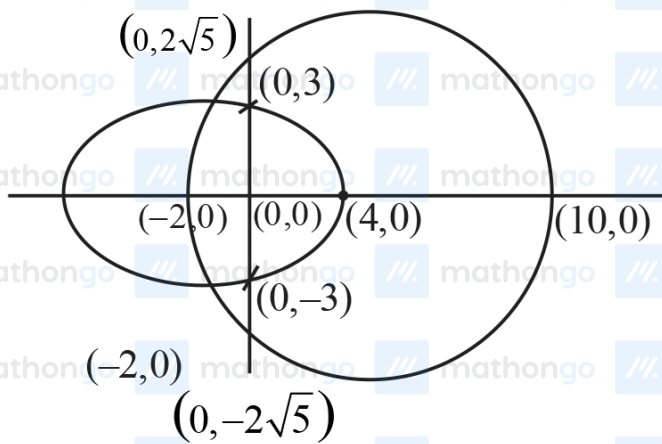
$$S: \frac{(x-3)^2}{16} + \frac{(y-4)^2}{9} \leq 1 ; x, y \in \{1, 2, 3, \dots\}$$

$$T: (x-7)^2 + (y-4)^2 \leq 36, x, y \in \mathbb{R}$$

Let $x-3 = x : y-4 = y$

$$S: \frac{x^2}{16} + \frac{y^2}{9} \leq 1 ; x \in \{-2, -1, 0, 1, \dots\}$$

$$T: (x-4)^2 + y^2 \leq 36 ; y \in \{-3, -2, -1, 0, \dots\}$$



$$S \cap T = (-2, 0), (-1, 0), \dots, (4, 0) \rightarrow (7)$$

$$(-1, 1), (0, 1), \dots, (3, 1) \rightarrow (5)$$

$$(-1, -1), (0, -1), \dots, (3, -1) \rightarrow (5)$$

$$(-1, 2), (0, 2), (1, 2), (2, 2) \rightarrow (4)$$

$$(-1, -2), (0, -2), (1, -2), (2, -2) \rightarrow (4)$$

$$(0, 3), (0, -3) \rightarrow (2)$$